HOYA 4

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Machine Learning II Project 3:
Targeted Travel Package Development: Customer Segmentation and Strategic Recommendations for Travelbiz's European Market

Hoya Consulting Group

Executive Summary

The marketing and product management team at Travelbiz wants to identify specific customer segments to target and the types of travel packages to develop for these segments, as they create and promote travel packages to Europe. To achieve this, Travelbiz has engaged the Hoya Consulting Group to analyze user behavior and preferences through available data, providing valuable recommendations for marketing strategies to target these customers.

To identify the specific customer segments to target along with the types of travel packages to develop for these segments, the Hoya Consulting Group leveraged two different techniques: Principal Component Analysis (PCA) and k-means clustering. Ultimately, four distinct traveler groups emerged based on their travel preferences, which can be targeted by specific offerings such as **The Urban Traveler Package**, **The Wellness and Relaxation Package**, **The Culture and Nature Explorer Package**, and the **Local Immersion Package**.

Hoya Consulting Group proposes the following recommendations for Travelbiz as they create and promote travel packages to Europe: (1) partner with local establishments such as restaurants, theaters, bars, gyms, etc. to provide bundles and discounts to customers, (2) offer travel packages specifically tailored to each distinct traveler group, and (3) target travel groups with tailored advertising on social media platforms and search engines.

Methodology and Data

Data Preprocessing

To prepare the data for modeling, the Hoya Consulting Group made several modifications to the original dataset. The dataset includes a row for each unique reviewer (n=5456) and columns representing each user's reviews of attractions across 24 categories in Europe, such as resorts, churches, beaches, malls, and restaurants. User ratings range from 1 to 5, and the average rating per category was calculated for each user.

During our data preprocessing, we identified a minimal proportion of missing values (NA) in the user rating columns. Given the negligible impact of these missing entries on our overall dataset, we opted to remove the rows containing NA values.

In the dataset, user ratings of zero could indicate either that the user did not visit that attraction category or that they visited and had no interest in submitting a review. The

proportion of zero ratings by category is illustrated in Figure 1 (see Appendix). For instance, categories such as 'Gym' have a high number of zero ratings, likely because many travelers may not be interested in using gym facilities during their vacation. The Hoya Consulting Group decided to keep user ratings of zero in the dataset. This decision was based on the understanding that a zero rating could signify that a person visited a destination but had a lack of interest in submitting a review. Because not leaving a review is significantly different from leaving a 1 star review, and we feel that it is very common to visit a place and have an enjoyable experience without leaving a review, we included all zero ratings in the dataset.

Finally, for both PCA and clustering, the user rating columns in the dataset were standardized. The standardization ensures that all variables are in the same scale as these techniques are sensitive to the variance of the original variables. Standardizing is a crucial step to ensure that PCA and clustering are performed optimally. After the data pre-processing steps, the data was prepared for running the following analysis techniques.

Principal Component Analysis

PCA is a statistical technique used to reduce the dimensionality of high-dimensional data while preserving as much variance as possible. By transforming the data into a new coordinate system, PCA identifies the directions (principal components) that capture the most significant variances in the data. The first principal component accounts for the largest variance, the second for the second largest, and so on. This process simplifies the data, making it easier to analyze and visualize without losing critical information.

PCA involves several steps: standardizing the data, computing the covariance matrix, calculating the eigenvalues and eigenvectors, sorting them in descending order of eigenvalues, and finally, projecting the original data onto the new feature space defined by the principal components. This technique is widely used for data visualization, noise reduction, feature extraction, and preprocessing for other machine learning algorithms. While PCA is effective in simplifying data, it has limitations, such as its linear nature and sensitivity to data scaling. Despite these challenges, PCA remains a powerful tool for uncovering patterns and relationships in complex datasets and identifying correlated variables.

Clustering Method

Clustering is a technique used to group a set of objects into clusters, where objects within the same cluster are more similar to each other than to those in other clusters. It relies on similarity or dissimilarity measures, often calculated using distance metrics.

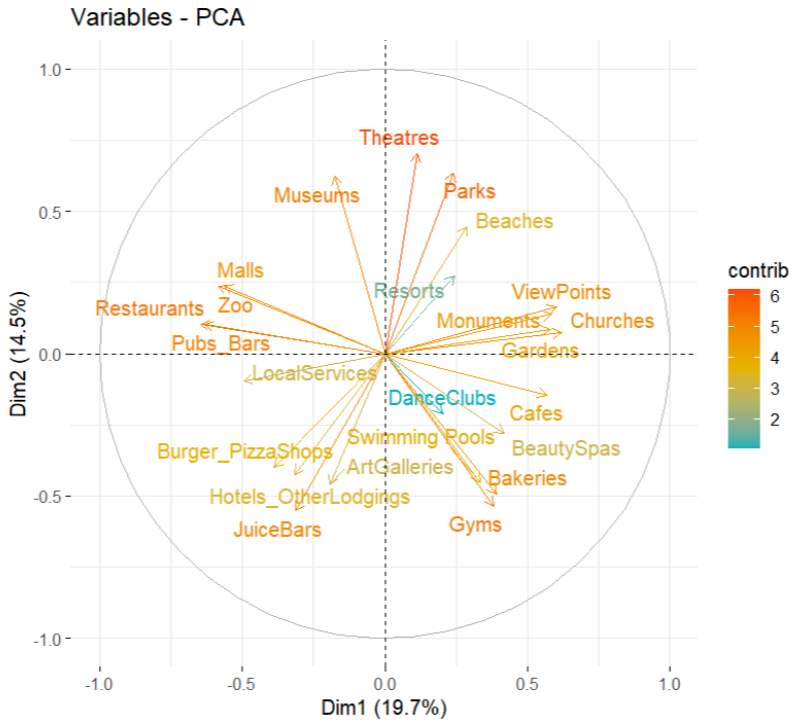
Clustering helps uncover hidden patterns and structures in data, facilitating data summarization and visualization. However, clustering has limitations, such as the subjective choice of the number of clusters and sensitivity to data scaling and distance metrics. Despite these challenges, clustering remains a powerful tool for exploring and understanding complex datasets, providing valuable insights for decision-making.

Key Findings

Principal Component Analysis

In PCA, Dimension 1 (Principal Component 1) and Dimension 2 (Principal Component 2) are ways of grouping clusters of users whose behavioral patterns were captured through data available on e-commerce and tourism-related platforms. Instead of looking at each destination amenity such as museums, resorts, and restaurants individually, the two dimensions illustrate the main types of travelers based on their preferences. Dimension 1 was found to explain 19.7% of the variance in traveler preference and Dimension 2 explained 14.5%, for a cumulative 34.2%.

Figure 2: Principal Component Analysis Variables by Dimension 1 and Dimension 2



As you can see in Figure 2, Dimension 1 or Principal Component 1 (PC1) is on the x-axis and Dimension 2 or Principal Component 2 (PC2) is on the y-axis, which splits the PCA plot into 4 quadrants with a vector for each destination amenity. A longer arrow or vector for a destination amenity with a higher contribution score (colored red) indicates that it contributes more to the variance in that principal component. When destination amenities appear in opposite quadrants, it suggests that these amenities likely serve a different consumer demographic, operate at different times, or serve different needs. In other words, the customer behavior is fundamentally different. When vectors are very close together (i.e. monuments, viewpoints, gardens), it indicates that these destination amenities serve similar types of travelers.

In the PCA plot, distinct consumer behavioral patterns emerge along the two principal components. PC1 differentiates the **Cultural and Nature-Based Traveler** who frequents monuments, gardens, churches, and viewpoints from the **Social and Urban Traveler** who gravitates toward malls, restaurants, pubs, and similar venues. PC2 further refines these distinctions, contrasting the **Outdoor and Daytime Activity Focused Traveler** who visits museums, theaters, parks, and beaches with the **Indoor and Service-Oriented Traveler**, who prefers amenities like juice bars, gyms, spas, and cafes.

Notably, within PC2, a further split emerges between two distinct traveler groups. The bottom left quadrant (burger/pizza shops, hotels, art galleries, and juice bars) reflects a preference for casual dining, accommodations, and cultural exploration. In contrast, the bottom right quadrant (cafes, beauty spas, swimming pools, bakeries, gyms, and dance clubs) represents wellness-conscious travelers who prioritize self-care, fitness, and nightlife. This suggests that those drawn to spas, gyms, and social nightlife exhibit different behaviors than those who favor casual eateries and lodging, highlighting the diverse preferences within the **Indoor and Service-Oriented Traveler** category.

Overall, the PCA analysis revealed 4 distinct traveler segments, including the **Cultural and Nature-Based Traveler**, **Social and Urban Traveler**, **Outdoor and Daytime Activity Focused Traveler**, and **Indoor and Service-Oriented Traveler**, which provides valuable insights for tourism operators who can tailor custom packages to these market segments.

Clustering Method

Using k-means clustering, different travelers can be grouped together based on their ratings across different activities and attractions. For example, some travelers might prefer visiting historical sites while others are more interested in local restaurants and bars, and they would be sorted into different groups. The k-means clustering algorithm continues to refine these groups until it finds the best way to organize travelers so that the people in each group are as similar to one another as possible and distinct from the other groups.

In order to choose the number of groupings, Hoya Consulting Group used two different analytical methods: the elbow method and the gap statistic method. While the elbow method showed that 4-6 distinct traveler groups would capture meaningful differences among travelers, the gap statistic method is more specific and indicated that 4 groups would be optimal. Additionally, a silhouette plot (see Appendix Figure 4), which shows how well travelers fit into their assigned groups, showed that most travelers have positive values, meaning they fit better in their assigned groups than other groups. A small number of travelers had negative values, meaning they are on the border between groups. The average silhouette width of 0.14 is relatively low (silhouette values range from -1 to 1), indicating that while the clusters are valid, they are not perfectly separated.

Figure 3, "Cluster Profiles for K-Means," illustrates four distinct traveler groups and their preferred destination amenities. Positive bars indicate a preference for a particular amenity, while negative bars suggest lower interest in that amenity:

1. Cluster 1 (Red) – **The Urban Traveler**:
   Shows high preference for restaurants, pubs and bars, zoos, museums, and local services, but lower ratings for other food-related venues like bakeries, cafes, and burger/pizza shops. They seek out an urban nightlife experience and cultural attractions.
2. Cluster 2 (Green) – **The Wellness & Lifestyle Seeker:**
   Shows high preference for bakeries, beauty spas, cafes, churches, dance clubs, gardens, swimming pools, gyms, and monuments, but lower ratings for pubs and bars, restaurants, and other local services. They strike a balance of relaxation, social spaces, and sight seeing with an eye for beauty.

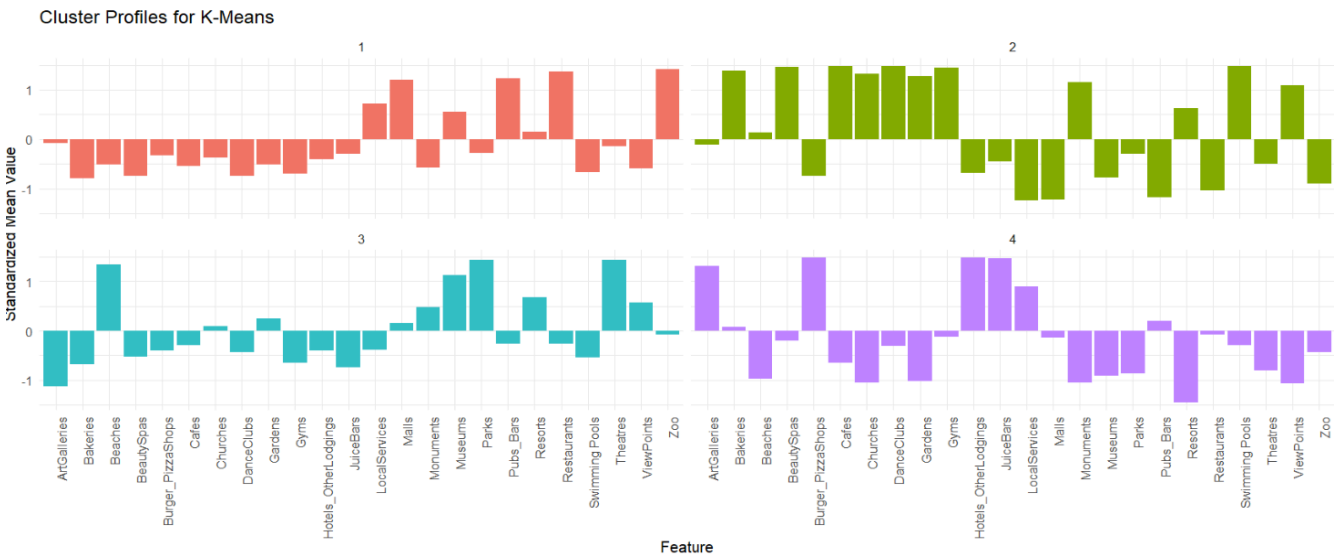
3. Cluster 3 (Turquoise): **The Nature and Cultural Enthusiast**:

   Shows high preference for theaters, parks, museums, and beaches, but lower ratings for food related attractions like cafes, bakeries, and restaurants and wellness attractions like gyms, swimming pools, and beauty spas. They enjoy cultural and outdoor experiences and are less drawn to dining and wellness amenities.

4. Cluster 4 (Purple): **The Local Explorer**:

   Shows high preference for art galleries, burger and pizza shops, hotels, juice bars, and other services, but lower preference for resorts, monuments, museums, beaches, churches, and gardens. Instead of tourist attractions, they seek out local gems and enjoy convenient dining and art experiences over outdoor activities and historical attractions.

Figure 3: Cluster Profiles for K-Means



Recommendations & Conclusion

Both the PCA and K-Means clustering models identified distinct traveler groups based on a diverse range of amenity preferences. First, Travelbiz should prioritize establishing partnerships with local businesses so that they are better able to establish discounts, promotions, and bundled experiences that can be provided to customers based on their specific preferences. These partnerships will enable Travelbiz to offer curated travel packages with unique perks that differentiate them from other travel businesses.

Travelbiz's management team should strategically tailor packages and promotions to align with distinct traveler groups, leveraging their preferences to maximize traveler engagement and appeal. Hoya Consulting Group recommends that Travelbiz promotes the following travel packages:

1. **The Urban Traveler Package** (**The Urban Traveler** | **The Social and Urban Traveler**)

- Target audience: young professionals, city explorers, foodies
- Key Recommendations:
    - Partner with local restaurants and bars to provide special discounts, food tours, wine tastings, etc.
    - Partner with zoos and museums to provide discounted entry and highlight special exhibits as they change seasonally
    - Offer public transportation packages so travelers can easily get around the city

2. **The Wellness and Relaxation Package** (**The Wellness & Lifestyle Seeker | Indoor and Service-Oriented Traveler**)

- Target audience: wellness enthusiasts, travelers looking for a relaxing and restorative vacation
- Key Recommendations:
    - Highlight resorts with spas, pools, and gyms
    - Provide a coffee and bakery walking tour that will highlight local favorites and includes the gardens along the way
    - Offer a ClassPass style workout package so that travelers can either check into a gym daily or choose from a variety of different classes each day, including yoga, pilates, boot camp style classes, and swimming
    - Offer wellness oriented experiences such as massages, facials, and pedicures

3. **The Culture and Nature Explorer Package (The Nature and Cultural Enthusiast | Outdoor and Daytime Activity Focused Traveler & Cultural and Nature-Based Traveler**

- Target audience: outdoor enthusiasts, history buffs
- Key Recommendations:

    - Partner with local theaters and promote discounted tickets to plays & musicals
    - Provide guided tours and discounted access to famous parks, museums, monuments, churches, gardens, and viewpoints
    - Provide a beach day package that allows travelers to easily rent chairs and towels
    - Offer surfing lessons

4. **The Local Immersion Package** (**The Local Explorer**)

- Target audience: travelers with an appreciation for locally curated experiences & trendy eateries, art aficionados
- Key Recommendations:
    - Promote boutique hotels that offer an authentic, local experience as compared to chain hotels
    - Provide food tours for pizza and burger shops that allow travelers to try multiple eateries in the same day
    - Partner with local art galleries, provide discounted tickets, and promote upcoming art exhibitions

Travelbiz should not only create travel packages tailored to each traveler group, but they should also leverage targeted social media advertising to reach the right audience. Based on social media users' likes, following, and browsing behavior, Travelbiz can strategically promote relevant packages for travelers that are most likely to be interested in them. For example, someone who follows pizza or burger chains on Instagram and likes posts from various artists would be a perfect target for ads selling the "Local Immersion" package.

In conclusion, by leveraging local partnerships, targeted travel packages, and data-driven social media advertising, Travelbiz can effectively engage distinct traveler groups and provide them with the best possible customer experience.

Appendix

Figure 1: Proportion of Zeroes By Category

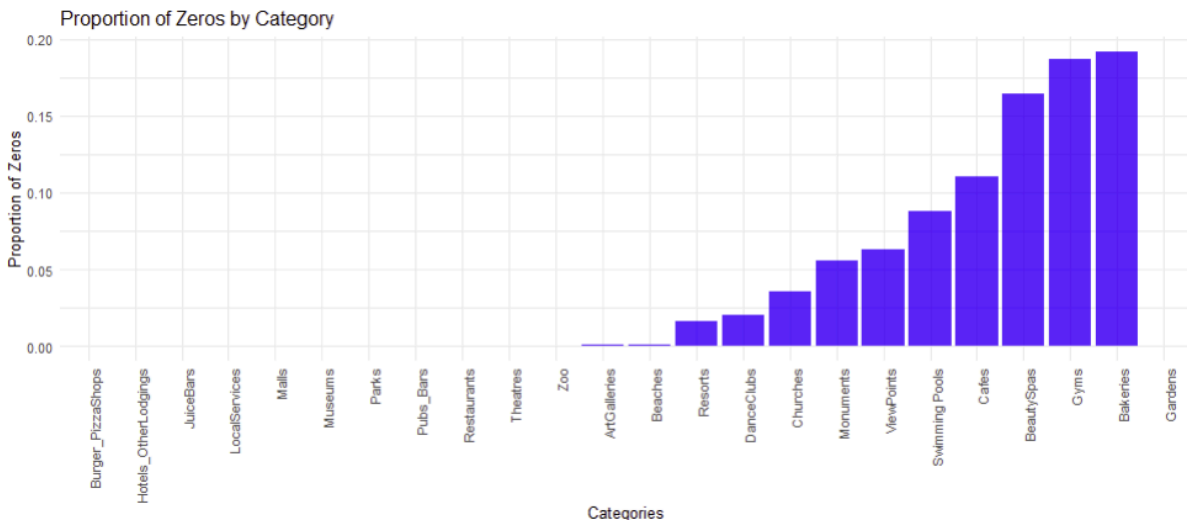
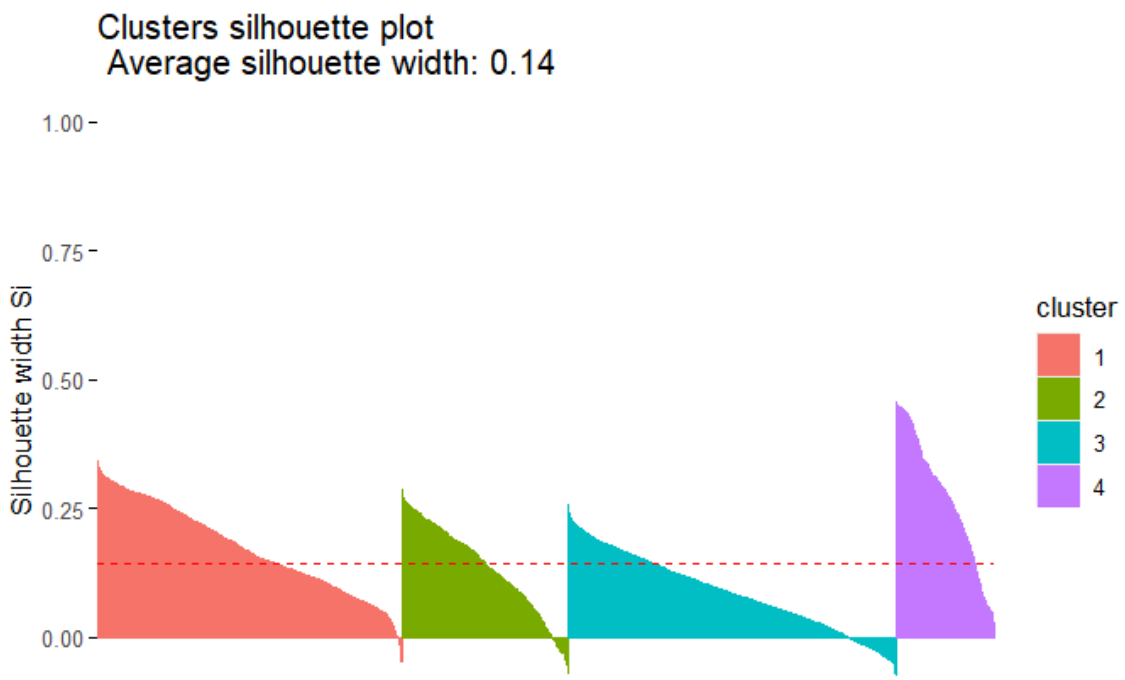



Figure 4: The Clusters Silhouette Plot